Aerospace Manufacturing Core
Study Guide

"AMCSG-103"

Introduction to Aerospace Manufacturing

Materials & Corrosion

Aircraft Materials

Text & Stats from Google AI

Primary Materials in Commercial Aircraft

Using newer **Boeing 787** or **Airbus A350** as a reference…

- 50% **Composites** (carbon fiber, epoxy, fiberglass, aramid fibers, …)
- 45% **Metals** (aluminum alloys, titanium, steel, …)
- 5% **Other** (plastics, wood, copper wiring, …)

Using traditional jetliners like **737** or **747**…

- 75-80% **Aluminum Alloys:** (most common alloys are 2024 & 7075)
- 10-15% **Steel**
- 5-10% **Titanium**
- < 1-3% **Other**

Source:

- List of Materials + Metrics: Google "what are the primary materials used in commercial jets today" >> AI tab … Added up 20% Aluminum + 15% titanium + 10% steel for metals

Text from Google AI

Material Properties

Materials are selected based on properties & performance

- **Strength:** Stronger is better ... four: tensile, shear, bearing, yield
- **Weight:** Lighter is better
- **Cost:** Lower cost is better

- **Durability:** Longer life is better ... corrosion is bad
- **Usability:** Easier maintenance is better
- **Appearance:** You know it when you see it ... avoid ugly

Text from Google AI

Properties of Aluminum Alloys

- Aluminum alloys **win** when
  - Ease of Fabrication (**Usability**) and (**Cost**) are critical
    (up to 80% by weight in traditional aircraft)
  - Fatigue and Corrosion Resistance (**Durability**) is critical

- Aluminum **is used in**:
  - **Fuselage & Wing Skins** for high strength, low weight, durability
  - **Structural** components for strength in wing ribs, stringers, etc.
  - **Other Components** like seat frames, cockpit/cabin furniture, etc.

(CC2) Wikimedia Commons

(Ad) Datron Milled Aluminum Part

- Dr. Google: “when is aluminum the best material to use on airplane”
- Dr. Google: “Aluminum in aircraft manufacturing process”

Image Credits:
- Google “B29” = CC2 - https://upload.wikimedia.org/wikipedia/commons/3/30/Boeing_B-29_Superfortress_%2819926000422%29.jpg
- Google “Picture of Datron milled aluminum part” = (Adv) https://www.datron.com/resources/blog/4-examples-of-milling-aerospace-parts/

Text from Google AI

Properties of Composites

- Composites **win** when:
  - **Cost**: Fuel Efficiency
  - **Cost**: Fewer Parts / Molded Shape
  - **Durability:** Fatigue and Corrosion Resistance

(News) Kawaski.com, 2019

- Composites **are used in**:
  - **Fuselage** & **Wing Skins** for high strength, low weight, durability
  - **Structural** components for strength in wing spars, stringers, etc.
  - **Fairings** for high impact resistance (aramid fibers)

(News) Composites World

- Dr. Google: "when is composites the best material to use on airplane"
- Dr. Google: "composites in aircraft manufacturing process"

Image Credits:

- Google "Kawasaki 1000th 787 Fuselage section" = (News) - https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20191216_8615
- Google "carbon fiber wing spar 777x" = (News) - https://theaircurrent.com/dispatches/beyond-the-carrier-deck-a-glimpse-at-the-777x-folding-wingtip/

Text from Google AI

Properties of Titanium

Titanium **wins** when:

- **Strength**: High Stress
- **Durability**: Temperature Resistance

- Titanium **is used in**:
  - **Engine** components for high heat
  - **Landing Gear & Hydraulics** for high fatigue resistance and strength
  - **Structural** airframes for strength in frames, splices, bolts, etc.

(CC3) Wikimedia Commons

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- Dr. Google: "when is titanium the best material to use on airplane"
- Dr. Google: "titanium alloy in aircraft manufacturing process"

Image Credits:

- Google: "picture of titanium bolts" = (CC0) - https://pxhere.com/en/photo/1208085
- Google: "777 landing gear wikimedia" = (CC3) - https://commons.wikimedia.org/wiki/File:Boeing-777-300_chassis.jpg

Text from Google AI

Properties of Fiberglass

Fiberglass **wins** when:

- **Usability**: Transparency to radar waves
- **Usability**: Ease of molding into complex aerodynamic shapes
- **Strength**: Impact resistance against bird-strikes, etc.
- **Durability**: Corrosion and fire resistance
- **Cost**: Much less than Carbon or Aramid fibers

(Ad) PlaneReclaimers

Fiberglass **is used in**:

- **Radomes** and **antennas** for radar and signal transparency
- Secondary structures like **engine cowlings**, **wingtips**, **fairings**, etc. for ease of molding and impact areas
- Interior components like **cabin walls**, **overhead bins**, **flooring**, etc. for high durability and low cost

(CC4) Wikimedia Commons

- Dr. Google: "when is fiberglass the best material to use on airplane"
- Dr. Google: "fiberglass in aircraft manufacturing process"

Image Credits:

- Google "Radome for sale": (Ad) - https://planereclaimers.aero/product/boeing-aircraft-radome/
- Google "inside 737 boeing.com": (CC4) - https://commons.wikimedia.org/wiki/File:WestJet_Boeing_737_MAX_8_Interior.jpg

Text from Google AI

Properties of Plastics

- Plastics **win** when
  - **Weight & Cost**: Very light & inexpensive
  - **Durability**: Maximum lifespan
  - **Usability**: Ease of manufacturing

Plastics **are used in**:

- **Luggage Bins & Panels** where low weight, low cost and reasonable strength is critical
- **Tray tables**, **armrests**, and **seat backs** for comfort are critical

(C) John Walton

(Ad) Dretloh Aircraft Supply

(Ad) Dretloh Aircraft Supply

- Dr. Google: "when is plastic the best material to use on airplane"
- Dr. Google: "plastic in aircraft manufacturing process"

Image Credits:

- Google "Picture of plastic overhead bins in 777" = (C) - https://runwaygirlnetwork.com/wp-content/uploads/2023/06/777X-header.-Image-John-Walton.jpg
- Google "plastic seat tray" = (Ad) - https://dretloh.com/wp-content/uploads/2024/08/Traytable-e1488995091768-1024x893.jpg
- Google "seat armrest" = (Ad) - https://dretloh.com/wp-content/uploads/2024/08/13430-31-URAL-Brice-7.jpg

Aircraft Corrosion

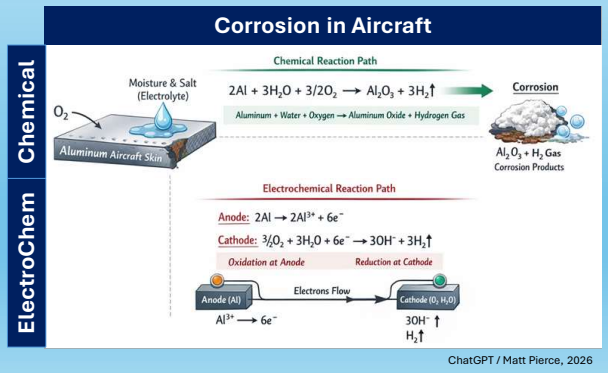
What is Corrosion?

"Corrosion turns metal to its natural state, a useless powder"

- **Example:** Rust

(PD) PxHere

- **How:**
  1. **Chemical Reaction Path (Slow):**
     Metal
     + Water
     + Oxygen
     = **Corrosion**
     ** Note: Thin, self-limiting layer on aluminum stops
  2. **Electrochemical Reaction Path (Fast):**
     Metal
     + Different Metal (far apart on galvanic corrosion chart)
     + Moisture / Oxygen / Acids / Bases / Salts / Oils
     = **Corrosion**



ChatGPT / Matt Pierce, 2026

Image Credits:

- Rusty carcass = (CC0) - https://pxhere.com/en/photo/1106074
- Chat GPT diagram = "diagram electrochemical reaction leading to corrosion in aircraft. show both chemical reaction and electrochemical reaction paths"

Text from Google AI

Types of Corrosion (1 of 4)

- **Uniform Corrosion** (Etching / Surface)
  - **Cause**: Direct exposure oxygen + chemicals at surface after paint wears off
  - **Appearance**: General dulling
  - Most common; low risk of catastrophic failure

ChatGPT / Matt Pierce, 2026

- **Pitting Corrosion**
  - **Cause**: Damaged oxide film + damaged paint + non-metallic inclusions
  - **Appearance**: White/gray powder; localized holes or cavities after clean
  - Hard to detect, prevent, design against
  - High risk of catastrophic failure; especially magnesium & aluminum

ChatGPT / Matt Pierce, 2026

- Excellent Source: https://www.ampp.org/technical-research/what-is-corrosion/forms-of-corrosion
- Google AI: “aircraft corrosion types”

Image Credits:

- Uniform Corrosion: ChatGPT “make a diagram showing uniform corrosion on a commercial aircraft part” + “Make better photo images”
- Pitting Corrosion: ChatGPT “make a photo showing pitting corrosion on a commercial aircraft part”

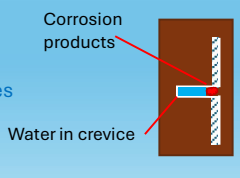
Text from Google AI

Types of Corrosion (2 of 4)

- **Galvanic Corrosion**
  - **Cause**: Two dissimilar metals touch; anode side corrodes rapidly
  - **Appearance**: White, powdery or chalking deposit with pitting or flaking paint
  - In AI picture, Aluminum aircraft skin is fastened with Stainless-Steel rivets

ChatGPT / Matt Pierce, 2026

- **Concentration Cell Corrosion** (Crevice)
  - **Cause**: Tight gap traps moisture & pollutants; different concentrations / chemistry in crevice
  - **Appearance**: Surface discoloration; at nameplates, riveted heads, lapped joints, etc.
  - May not be visible until failure
  - In AI picture, water and debri accumulate in crevices



ChatGPT / Matt Pierce, 2026

- Excellent Source: https://www.ampp.org/technical-research/what-is-corrosion/forms-of-corrosion
- Google AI: “aircraft corrosion types”
- Galvanic Corrosion: ChatGPT “make a photo showing galvanic corrosion on a commercial aircraft skin with stainless steel rivets”
- Crevice Corrosion: ChatGPT “make a photo showing crevice corrosion on a commercial aircraft skin”

Text from Google AI

Types of Corrosion (3 of 4)

- **Filiform Corrosion**
  - **Cause:** Humidity 78 to 90; surface slightly acidic; starts at breaks in protective coating
  - **Appearance:** Worm-like trace under painted or plated surfaces
  - Special form of "Crevice Corrosion"
  - Lacquers and quick-dry paints most susceptible
  - If left untreated, leads to intergranular corrosion

ChatGPT / Matt Pierce, 2026

- **Fretting Corrosion**
  - **Cause**: Debris between two tight vibrating surfaces = abrasive and corrosive
  - **Appearance**: Difficult to spot until late; pits and black debris
  - Located at bolted joints, splined shafts, bearings, and other clamped surfaces prone to tiny rubbing

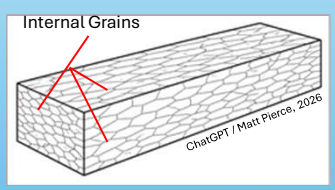
ChatGPT / Matt Pierce, 2026

- Excellent Source: https://www.ampp.org/technical-research/what-is-corrosion/forms-of-corrosion
- Google AI: "aircraft corrosion types"
- Filiform Corrosion: ChatGPT "make a photo showing filiform corrosion on a commercial aircraft skin"
  + "make the rust spider veins silver color and make them bubble out more from the skin"

Text from Google AI

Types of Corrosion (4 of 4)

- **Intergranular Corrosion** (Exfoliation)
  - **Levelset**: Microstructure of metals and alloys comprised of grains
  - **Cause:** Corrosion follows weak internal grain boundaries deep inside
  - **Appearance**: Difficult to spot until late exfoliation begins
  - Extremely dangerous / major hazard of high-strength aluminum alloys
  - Heat treating changes grain structure inside part
  - Requires ultrasonic or x-ray inspection to detect



ChatGPT / Matt Pierce, 2026

- Excellent Source: https://www.ampp.org/technical-research/what-is-corrosion/forms-of-corrosion
- Google AI: “aircraft corrosion types”
- Intergranular Corrosion: ChatGPT “make a photo showing “intergranular” corrosion following grains under a magnifying glass on a high-strength aluminum alloy part of a commercial aircraft”
- Aluminum Bar with Intergranular grains: ChatGPT: “make a diagram showing a 2" x 4" x 4' aluminum bar. Add 20 randomly patterned internal grains on each of the three surfaces. Label them "short longitudinal", "short transverse", and "longitudinal transverse".”
  + “Convert this into a single clean presentation-ready image. Redraw it as an ASTM-style metallography diagram.”
  + “Remove the dimensions. Remove titles. Only keep the longer bar.”
  + “Remove titles “short longitudinal” and “short transverse and fix the bottom corner to tie in all three surfaces”
  + “make the longitudinal sides have grains that are 5 time longer. Leave the short face grains the same size.”
- Fretting Corrosion: ChatGPT “make a photo showing fretting corrosion between commercial airliner aluminum skin and the aluminum rivets vibrating

against it. Show rust stains below each rivet.”
+ “reduce the corrosion by half. reduce the number of fasteners by half”

Text from Google AI

Corrosion Prevention

- Use of **coatings** to protect metals
  - **Anodizing** for Aluminum
  - **Zinc-** or **Cadmium-plating** for Steel
  - **Special Treatments** for Magnesium and Titanium
    - Magnesium is most anodic metal – highly susceptible to corrosion
    - Titanium is most cathodic metal – highly corrosion resistant, but contact will corrode other metals
- Use of more **corrosion-resistant materials**
- Improved **adhesive bonding** processes
- Use of **sealants in fastener holes** and **faying surfaces**
- **Control of fluid spillage** in galleys and lavatories

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Text from Google AI

Corrosion Prone Areas

- **Exhaust Trail** Area behind engines (highly corrosive gases)
- **Battery Compartment** Areas (highly corrosive fumes, needs venting)
- **Lavatories, Buffets, and Galleys** (spilled water, food, and garbage)
- **Bilge** Areas (designed to collect waste, fluids, dirt, loose fasteners, paint chips…)
- **Wheel Wells** and **Landing Gear** (high exposure to elements: mud, water, salt, gravels, whatever comes up from runway)

- Excellent Source: https://www.ampp.org/technical-research/what-is-corrosion/forms-of-corrosion
- Google AI: “aircraft corrosion types”
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Q&A

Practice Q&A's (Q1-Q8)

Q1: What are currently (2026) the two most common materials used in aircraft?

A: Aluminum alloys and composites

Q2: What are the most common metals used in aircraft?

A: Aluminum, titanium and steel

Q3: What are the most common aluminum alloys used in aircraft?

A: 2024 and 7075

Q4: What are the three most important material properties in aircraft design?

A: Strength, weight, and cost

Q5: What are three secondary material properties that also factor into aircraft design?

A: Durability, usability, and appearance

Q6: What material is best for ease of fabrication (usability), cost, and fatigue/corrosion resistance (durability)?

A: Aluminum

Q7: Aluminum can be used in skins, structural components (ribs, stringers), seat frames, etc.?

A: True

Q8:What material is best for fuel efficiency, simplicity (fewer parts), and high fatigue and corrosion resistance (durability)?

A: Composites

Practice Q&A's (Q9-Q16)

Q9: Composites can be used in skins and structural components (spars, stringers, etc.)?

A: True

Q10: What material is best for high strength, high stress, and temperature resistance?

A: Titanium

Q11: Titanium can be used in engine parts, landing gear, hydraulics, and structural airframes for high strength splices and bolts?

A: True

Q12: Fiberglass is used in radomes, antennas, engine cowlings, and wingtips.

A: True

Q13: What material is best for transparency to radar waves, impact resistance against bird strikes, is corrosion and fire resistant, and is the lower cost?

A: Fiberglass

Q14: What material is best for very lightweight, very inexpensive, very easy to manufacture, very long-lasting, but least strong ?

A: Plastic

Q15: Plastic can be used for structural elements (spars, strings, etc.) and skins?

A: False

Q16: Chemical corrosion can occur slowly with metal + water + oxygen.

A: True

Practice Q&A's (Q17-Q24)

Q17: Chemical corrosion can occur with metal + acid

A: True

Q18: What is another name for electrochemical corrosion?

A: Galvanic corrosion

Q19: What is required to make galvanic corrosion?

A: Two different metals conducting electrical charge thru moisture, acid, base, salt, or oil

Q20: What type of corrosion is the most common, has a general dulling, and is low risk?

A: Uniform corrosion

Q21: What type of corrosion looks like cavities after cleaned, is hard to detect, prevent and design against, and is high risk?

A: Pitting corrosion

Q22: What type of corrosion involves two dissimilar metals touching, where anode side corrodes rapidly?

A: Galvanic corrosion

Q23: What type of corrosion occurs at tight gaps of the same overlapping material?

A: Concentration cell corrosion

Q24: What type of corrosion occurs at humidity 78 to 90 and looks like work-likes tunnels?

A: Filiform corrosion (aka crevice corrosion)

Practice Q&A's (Q25-Q32)

Q25: What type of corrosion occurs between two tight vibrating surfaces (abrasive & corrosive)?

A: Fretting corrosion

Q26: What type of corrosion occurs microscopically at the grains inside the allow (like wood grains), and is extremely dangerous?

A: Intergranular corrosion

Q27: Anodizing is a process to protect aluminum from corrosion.

A: True

Q28: Zinc- or Cadmium-plating is a process to protect steel from corrosion.

A: True

Q29: Using sealants at joints and penetrations protects metal from corrosion.

A: True

Q30: What exterior facing surfaces are prone to corrosion?

A: Wheel wells, landing gear, and exhaust trail (behind engines)...anywhere dirt, debris and hot chemicals accumulate

Q31: What interior facing surfaces are prone to corrosion?

A: Battery compartment, lavatories, buffets, galleys, and bilge areas...anywhere water can collect

Q32: What is the biggest threat to metal on an aircraft?

A: Corrosion

Practice Q&A's (Q33-Q37)

Q33: What are plastics primarily used for?

A: Interior cabin

Q34: What material is lightest and strongest?

A: Composites (esp. carbon fiber)

Q35: What is the primary disadvantage to using aluminum components?

A: Labor cost to assemble and rivet or bolt it all together.

Q36: What type of metal is a combination of two or more pure metals into a single new stronger, better metal?

A: Alloy

Q37: What is a more important criteria than strength, weight, cost, durability, usability, and appearance?

A: Safety!!!

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg